

Local Orbit Consistency for Symmetry-Robust Neural Operators

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Abstract

Neural operators can learn accurate PDE solution maps while still violating known symmetries of the underlying equations. We study stochastic local orbit consistency, a training objective that compares the prediction on a transformed input with the transformed original prediction and requires only model evaluations and known group actions. It requires no architectural change, supports labeled training and unlabeled adaptation, and leaves inference identical to the base operator. In a periodic N64 2D Navier–Stokes Galilean setting, augmentation plus orbit consistency reduces symmetry-induced OOD error and equivariance defect relative to supervised and augmentation baselines, with the strongest gains in low-label regimes. A five-seed reproduction with a near-capacity-matched DeepONet lowers OOD error by 7.0% and equivariance defect by 29.6% relative to its augmentation baseline, supporting portability to a second operator family while revealing backbone-dependent effect size.

1 Introduction

Neural operators are attractive for PDE learning because they approximate maps between function spaces rather than single finite-dimensional predictions. This flexibility also exposes a weakness: a high-accuracy operator need not respect symmetries that the exact solution operator satisfies. When test inputs move along a known symmetry orbit, such as a Galilean boost of a periodic flow, ordinary supervised training can produce both higher prediction error and larger equivariance defect.

The practical motivation extends beyond benchmark PDEs. Global atmospheric forecasting operates on spherical fields, where flat longitude–latitude operators can introduce coordinate artifacts and geometry-aware operators improve stability [3]. Neural operators are also used for vehicle-surface pressure and drag [17], frame-consistent tensorial turbulence closures [11], and plasma surrogates trained on simulated MHD and experimental tokamak observations [10]. At atomistic scale, forces must co-rotate with molecular coordinates, motivating $E(3)$ -equivariant interatomic models [2]. These are motivations rather than applications established here: an action is admissible only when geometry, state, boundary and forcing data, diagnostic coordinates, and vector or tensor outputs transform jointly. Fixed geography, walls, or observation geometry may reduce or break a nominal symmetry.

We propose to reduce this failure mode with local orbit consistency. Given a known group action on inputs and outputs, the objective samples a small finite transform and penalizes the discrepancy between $\mathcal{G}_\theta(T_g a)$ and $T_g \mathcal{G}_\theta(a)$. The penalty is a training-time regularizer, not a new inference pipeline. This makes it complementary to supervised Lie augmentation and architecture-level equivariance: it can be added to an existing FNO, applied in low-label settings, and evaluated on unlabeled inputs.

Our empirical claim is deliberately narrow. The strongest evidence is the main N64 2D Galilean experiment, where augmentation plus orbit consistency with the same FNO2d backbone lowers orbit OOD relative L^2 and equivariance defect compared with forward-evaluation-matched supervised augmentation. A near-capacity-matched, fixed-grid DeepONet reproduction finds the same direction on that benchmark but a smaller predictive-error reduction. Burgers and D4 provide supporting evidence in other symmetry settings, and a fixed-molecule MLP force predictor provides a secondary mechanism check beyond Fourier layers; none establishes uniform gains across arbitrary equations or architectures. The core N64 artifact path, manifest, tables, and figures are regenerated by `scripts/reproduce_paper_artifacts.py`; reviewer-control tables use the separately documented `scripts/run_reviewer_followup_experiments.ps1` workflow.

2 Related Work

Neural operators learn mappings between function spaces and are therefore a natural model class for parametric PDE solution operators. DeepONet introduced a branch-trunk architecture for learning nonlinear operators from function samples and output coordinates [19]. Fourier Neural Operators (FNOs) parameterize global integral kernels in Fourier space and provide an efficient backbone for PDE families on regular grids [15]. The general neural-operator framework formalizes these models as discretization-invariant compositions of integral operators and nonlinearities, with applications to Burgers’ equation, Darcy flow, and Navier–Stokes [13]. Physics-Informed Neural Operators (PINOs) add PDE residual constraints to operator learning, combining supervised operator regression with physics-based optimization [18]. Subsequent work extends neural operators to geometric settings: Geo-FNO learns deformations from irregular physical domains to latent uniform grids [16], GINO combines graph and Fourier operator components for large-scale three-dimensional PDEs on varying geometries [17], and NORM uses Laplacian eigenfunctions to define neural operators on Riemannian manifolds [5]. Operator-learning benchmarks such as PDEBench and The Well provide reusable datasets and baselines for time-dependent PDEs and broader physical simulation families [20, 23]. This literature establishes FNO-style models as computationally efficient solution-operator backbones, while also showing that domain geometry, discontinuities, and discretization effects require additional structure [14].

Symmetry has been used as an inductive bias through architectures, losses, and data transformations. Group-equivariant convolutional networks replace ordinary convolution by group convolution to share weights across symmetry orbits [8]; steerable CNNs generalize this principle through representation-constrained kernels [25]; and gauge-equivariant CNNs extend equivariance from global transformations to local frame changes on manifolds [9]. For operator learning, Group-Equivariant FNOs encode translation, rotation, and reflection symmetries in Fourier layers [12]. A parallel line derives symmetries from the governing differential equations. Lie Point Symmetry Data Augmentation constructs solution-preserving transformations from a PDE’s Lie point symmetry group and uses them to reduce sample complexity for neural PDE solvers [4]. Lie-symmetry losses for PINNs enforce conservation of solution families under infinitesimal generators [1]. Generalized Lie symmetries for PINOs further argue that evolutionary representatives can provide stronger training signals than point symmetries alone [24]. These works support the central premise that PDE symmetries should constrain the learned solution operator, not only the training data distribution.

Model-agnostic symmetry methods provide the closest precedent for the proposed direction. Tangent propagation used transformation tangents to impose local invariance constraints on neural networks [22]. Lie Algebra Canonicalization uses infinitesimal generators to construct canonicalized inputs for pretrained models and neural operators under arbitrary Lie groups, including non-compact groups commonly arising in PDEs [21]. PACE-FNO applies a related canonicalization strategy to FNOs under symmetry-induced shifts, estimating a Lie-algebraic input frame, predicting in a canonical frame, and restoring the output frame for translations, rotations, and Galilean boosts [26]. LOCO occupies a distinct point in this design space. Unlike supervised Lie augmentation, its consistency term needs no transformed label; unlike hard-equivariant layers, it does not restrict the backbone; unlike canonicalization, it does not alter inference; and unlike tangent propagation, it uses ordinary evaluations at finite group elements rather than an explicit Jacobian–vector product at the identity. Relative to a plain finite-transform consistency loss, the local $\epsilon^2 + \eta$ normalization removes the leading quadratic dependence of the loss scale on step magnitude. The revision tests the finite-versus-tangent distinction and reports a scale-matched unnormalized control rather than treating these objectives as interchangeable.

3 Method

We train a neural operator $\mathcal{G}_\theta : \mathcal{A} \rightarrow \mathcal{U}$ with ordinary supervised regression and a stochastic local orbit-consistency objective. Each minibatch contains input–solution pairs (a, u) . The supervised term is

$$\mathcal{L}_{\text{sup}}(\theta) = \mathbb{E}_{(a,u)} \left[\frac{\|\mathcal{G}_\theta(a) - u\|_2}{\max(\|u\|_2, 10^{-8})} \right]. \quad (1)$$

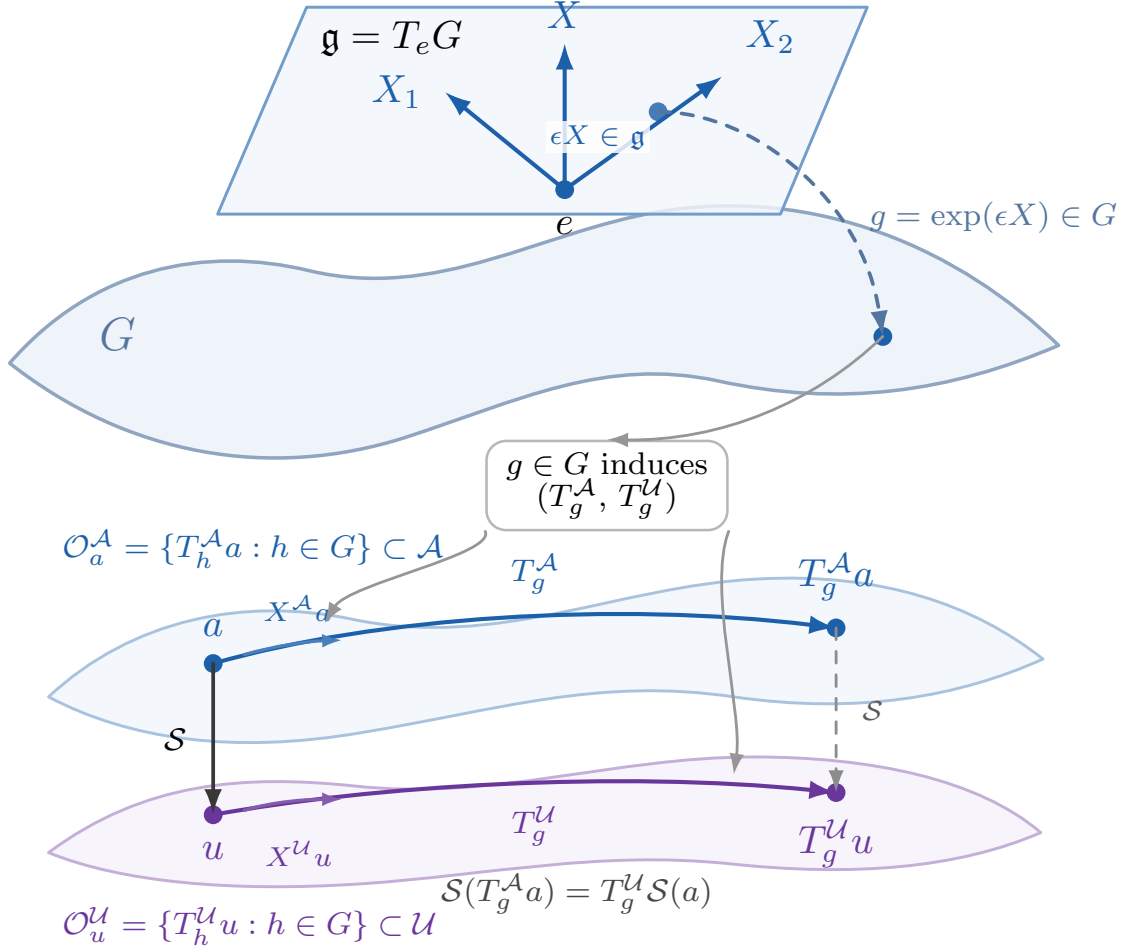


Figure 1: Lie-group actions on input and output field spaces. A local Lie-algebra step $\epsilon X \in \mathfrak{g} = T_e G$ maps through the exponential to a group element $g = \exp(\epsilon X) \in G$. The same g induces paired actions T_g^A and T_g^U on the input and output orbits. For an equivariant exact solution operator $\mathcal{S}$, the diagram commutes: $\mathcal{S}(T_g^A a) = T_g^U \mathcal{S}(a)$.

For a sampled Lie-algebra direction and finite transform g , let ϵ denote the step magnitude (for the two-dimensional Galilean action, $\epsilon = \max(\|\delta\|_2, 10^{-6})$). The implemented local orbit-consistency term is

$$\mathcal{L}_{\text{orb}}(\theta) = \mathbb{E}_{a,g} \left[\frac{\text{MSE}(\mathcal{G}_\theta(T_g^A a) - T_g^U \mathcal{G}_\theta(a))}{\epsilon^2 + \eta} \right], \quad (2)$$

where MSE averages over all spatial and output entries of each example before the batch average. Supervised Lie augmentation uses the same unsquared per-example relative loss,

$$\mathcal{L}_{\text{aug}}(\theta) = \mathbb{E}_{(a,u),g} \left[\frac{\|\mathcal{G}_\theta(T_g^A a) - T_g^U \mathcal{G}_\theta(u)\|_2}{\max(\|T_g^U u\|_2, 10^{-8})} \right]. \quad (3)$$

The combined method minimizes $\mathcal{L}_{\text{sup}} + \lambda_{\text{aug}} \mathcal{L}_{\text{aug}} + \lambda_{\text{orb}} \mathcal{L}_{\text{orb}}$; terms not used by a method are omitted.

What locality and step normalization add. A generic finite-transform consistency penalty would instead minimize $\mathbb{E}_{a,g}[\text{MSE}(\Delta_\theta(a, g))]$, without dividing by the local step. Near the identity, $\Delta_\theta(a, \exp(\epsilon X)) = \epsilon d_\theta(a; X) + O(\epsilon^2)$. The unnormalized penalty therefore scales as $\mathbb{E}[\epsilon^2 \|d_\theta\|^2] + O(\mathbb{E}[\epsilon^3])$: larger sampled steps

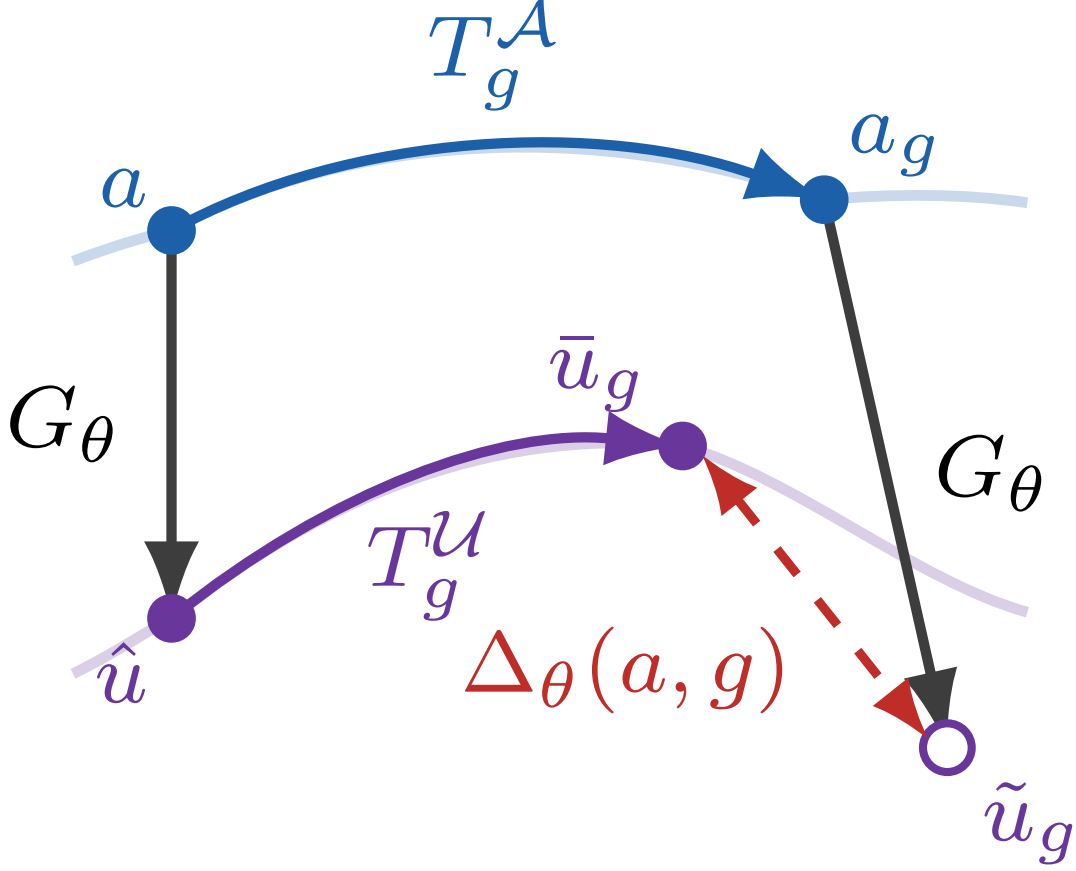


Figure 2: Local orbit consistency. For $g = \exp(\epsilon X)$, let $a_g = T_g^A a$, $\hat{u} = G_\theta(a)$, $\bar{u}_g = T_g^U \hat{u}$, and $\tilde{u}_g = G_\theta(a_g)$. LOCO penalizes the red equivariance defect $\Delta_\theta(a, g) = \tilde{u}_g - \bar{u}_g$, equivalently $\text{MSE}(\tilde{u}_g - \bar{u}_g)/(\epsilon^2 + \eta)$ in the discrete implementation.

receive quadratically greater leading-order weight, and the loss scale changes when the transform radius changes. Dividing by $\epsilon^2 + \eta$ removes this leading scale when $\epsilon^2 \gg \eta$, so the weight is interpretable against a tangent defect and is less mechanically coupled to the chosen local radius. The stabilizer caps the weight near zero. This normalization does not discard finite-orbit information: at nonzero steps the exact nonlinear group action retains higher-order terms that tangent propagation at the identity does not see. Section 6 compares the normalized objective with both explicit tangent propagation and a scale-matched unnormalized finite-transform control.

The loss is architecture-agnostic in the implementation sense: it requires only model evaluations and known input/output actions, not Fourier layers. We test it primarily with FNO backbones, reproduce the main Galilean comparison with a near-parameter-matched DeepONet, and separately use a non-equivariant molecular MLP as a mechanism check. This does not imply backbone-independent effect size. Transformations are applied to physical-grid input fields; the prediction on the transformed input is compared with the transformed original prediction; gradients pass through both model evaluations. Integer periodic translations are tensor rolls, non-integer periodic translations are spectral shifts, and finite square symmetries are exact tensor rotations and flips. For 1D Burgers, the Galilean action on the solution map $u_0 \mapsto u_T$ is $T_c^A u_0(x) = u_0(x) + c$ and $T_c^U u_T(x) = u_T(x - cT) + c$.

The same loss supports unlabeled adaptation. Given a pretrained operator $\mathcal{G}_{\theta_0}$ and target inputs $a \sim \mathcal{D}_{\text{target}}$, we minimize $\mathcal{L}_{\text{orb}}^{\text{target}}(\theta) + \beta \|\theta - \theta_0\|_2^2$ over a chosen adaptation set Θ_{adapt} , such as all parameters, the last operator block, or the projection head. This adaptation objective requires no PDE solves and no target labels.

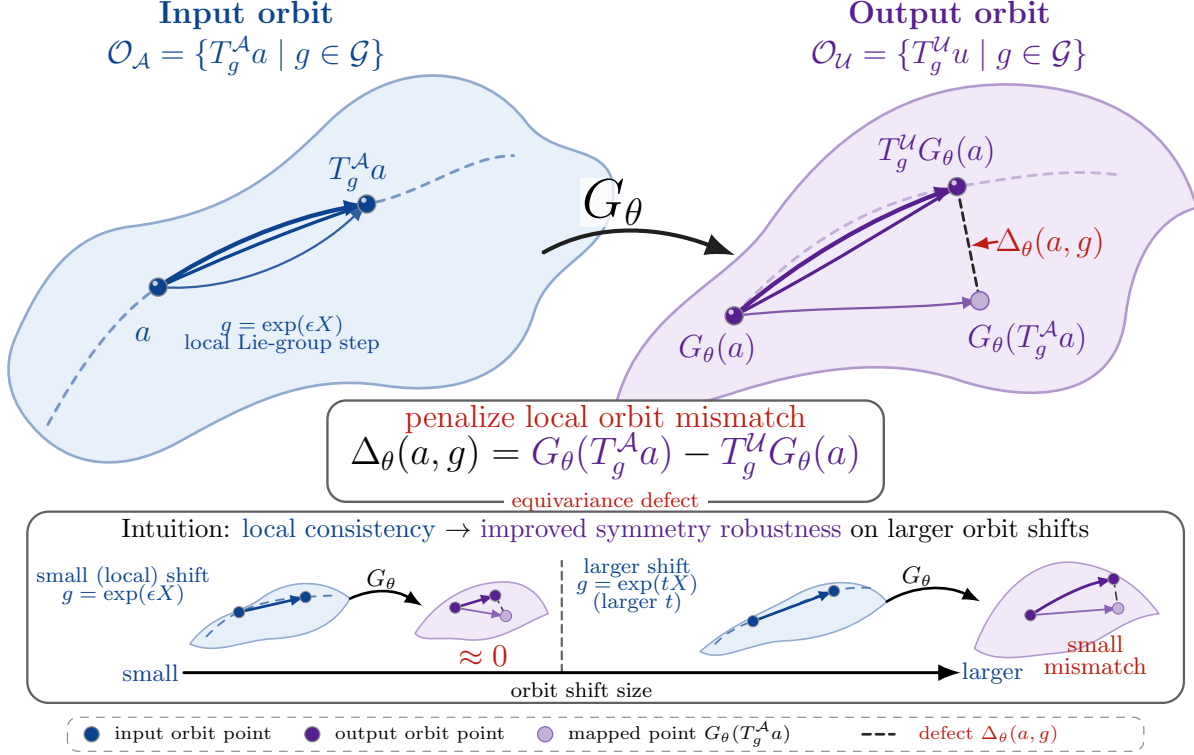


Figure 3: Vector schematic of local orbit consistency. A local Lie-group step moves an input a to $T_g^A a$. The training penalty compares the prediction on the transformed input, $G_\theta(T_g^A a)$, with the transformed original prediction, $T_g^U G_\theta(a)$, reducing the equivariance defect along the orbit while leaving inference unchanged.

4 Theory

Let $\mathcal{A}$ and $\mathcal{U}$ denote Banach spaces of input and output fields. A PDE family induces a solution operator $\mathcal{S} : \mathcal{A} \rightarrow \mathcal{U}$. Let G be a Lie group acting on the full problem family through transformations T_g^A and T_g^U . Closure of the full problem class under G means that the equation, coefficients, forcing terms, boundary or initial conditions, domain class, and observation map transform consistently. Under this assumption the exact solution operator is equivariant: $\mathcal{S}(T_g^A a) = T_g^U \mathcal{S}(a)$. For a learned operator $\mathcal{G}_\theta$, define the operator-level equivariance defect $\Delta_\theta(a, g) = \mathcal{G}_\theta(T_g^A a) - T_g^U \mathcal{G}_\theta(a)$. Our regularizer jointly samples a Lie-algebra direction X , a step ϵ , and $g = \exp(\epsilon X)$, then penalizes

$$\mathcal{L}_{\text{orb}}(\theta) = \mathbb{E}_{a, X, \epsilon} \left[\frac{\|\Delta_\theta(a, \exp(\epsilon X))\|_{\mathcal{U}}^2}{\epsilon^2 + \eta} \right]. \quad (4)$$

For gridded experiments, $\|\cdot\|_{\mathcal{U}}^2$ is discretized as the mean square over spatial and output entries, matching the implementation in Section 3. The objective is label-free because both terms are model predictions transformed by known group actions. For the N64 square sampling law, $X = \delta/\|\delta\|_2$ away from the measure-zero identity and $\epsilon = \|\delta\|_2$; direction and step are therefore jointly induced rather than independently sampled.

Infinitesimal interpretation. Assume $T_{\exp(\epsilon X)}^A$, $T_{\exp(\epsilon X)}^U$, and $\mathcal{G}_\theta$ are differentiable at $\epsilon = 0$. Let $X^A a$ and $X^U u$ denote the induced infinitesimal actions. Taylor expansion gives

$$\mathcal{G}_\theta(T_{\exp(\epsilon X)}^A a) - T_{\exp(\epsilon X)}^U \mathcal{G}_\theta(a) = \epsilon (D\mathcal{G}_\theta(a)[X^A a] - X^U \mathcal{G}_\theta(a)) + O(\epsilon^2). \quad (5)$$

Thus $\|\Delta_\theta(a, \exp(\epsilon X))\|_{\mathcal{U}}^2/\epsilon^2$ converges to $\|D\mathcal{G}_\theta(a)[X^A a] - X^U \mathcal{G}_\theta(a)\|_{\mathcal{U}}^2$ as $\epsilon \rightarrow 0$. For the implemented denominator $\epsilon^2 + \eta$, this tangent interpretation applies when $\epsilon^2 \gg \eta$, or along a limit with $\eta = o(\epsilon^2)$; with

fixed positive η , the loss instead vanishes as $\epsilon \rightarrow 0$. LOCO is therefore a finite-difference-based regularizer that avoids explicit Jacobian–vector products, not an unbiased tangent estimator at the finite radii used experimentally. For nonzero ϵ , the normalized defect also contains higher-order variation of the learned map and group orbit. Enforcing commutation over a sampled neighborhood can therefore constrain finite displacements that an identity-only directional derivative does not; the tangent and finite-step controls in Section 6 test this distinction.

Orbit error decomposition. If the true solution operator is equivariant and $T_g^{\mathcal{U}}$ is bounded, then for any $a \in \mathcal{A}$ and $g \in G$,

$$\begin{aligned} \|\mathcal{G}_\theta(T_g^{\mathcal{A}}a) - \mathcal{S}(T_g^{\mathcal{A}}a)\|_{\mathcal{U}} &\leq \|\mathcal{G}_\theta(T_g^{\mathcal{A}}a) - T_g^{\mathcal{U}}\mathcal{G}_\theta(a)\|_{\mathcal{U}} \\ &\quad + \|T_g^{\mathcal{U}}\| \|\mathcal{G}_\theta(a) - \mathcal{S}(a)\|_{\mathcal{U}}. \end{aligned} \quad (6)$$

The first term is the learned equivariance defect and the second term is the in-distribution prediction error transported by the output action. This inequality motivates reducing $\mathcal{L}_{\text{orb}}$ when test inputs differ from training inputs by known PDE symmetries.

Approximate symmetries and cost. For partial or approximate symmetries, replace the output norm by a weighted norm $\|v\|_w^2 = \|w_{a,g} \odot v\|^2$, where $w_{a,g}$ can encode valid overlap after transformations, boundary exclusions, observation masks, or regions where transformed coefficients and forcing terms remain physically meaningful. With one sampled transform per minibatch, the LOCO term uses one evaluation on the original input, one on the transformed input, and a known output transformation. At deployment there is no LOCO module or additional operator call: inference uses the same base graph $\mathcal{G}_\theta(a)$, and the measured latency differences are within run-to-run variation.

5 Experimental Details and Evaluation Protocol

This section gives the complete protocol behind the reported results. We distinguish three quantities that were conflated in the shorter submission: the ambient frame used to generate each Navier–Stokes example, the relative transform sampled during training, and the relative transform used for the symmetry-shifted test evaluation. All “radii” below are componentwise bounds. Thus $\delta \sim \mathcal{U}([-r, r]^2)$, not a uniform draw from a Euclidean ball, and the step used in the LOCO denominator is $\epsilon = \max(\|\delta\|_2, 10^{-6}) \leq \sqrt{2}r$. The clamp is a numerical guard for the tiny region $\|\delta\|_2 < 10^{-6}$, not only the exact identity.

5.1 Headline N64 Galilean experiment

Equation, domain, and numerical solver. We solve the unforced vorticity equation

$$\partial_t \omega + v \cdot \nabla \omega = \nu \Delta \omega, \quad v = (\partial_y \psi, -\partial_x \psi) + c, \quad \Delta \psi = \omega, \quad (7)$$

on the periodic square $[0, 1]^2$. This sign convention is the one used by the released solver. The solution is represented on a 64×64 uniform grid and advanced from $t = 0$ to $T = 0.5$ with a pseudo-spectral classical fourth-order Runge–Kutta method, time step 10^{-3} (500 steps), viscosity $\nu = 10^{-3}$, tensor-product 2/3-rule de-aliasing of the nonlinear term, and no forcing. Solver generation uses batches of eight only to limit memory; it does not change the numerical method.

Initial conditions and ambient frames. For every example, we draw independent standard Gaussian noise z on the grid, apply the Fourier filter $\hat{\omega}_0(k) = \hat{z}(k) \exp(-\|k\|_2^2/8)$, subtract the spatial mean, divide by the sample spatial standard deviation plus 10^{-6} , and use amplitude one. We then draw the constant ambient-velocity components independently as $c_x, c_y \sim \mathcal{U}[-0.5, 0.5]$. The input channels are $(\omega_0, c_x, c_y) \in \mathbb{R}^{64 \times 64 \times 3}$, with the two scalar velocities broadcast over the grid; the target is the terminal vorticity $\omega(T) \in \mathbb{R}^{64 \times 64 \times 1}$. All stored tensors are `float32`. We apply no model-side input or target normalization, discard no examples, and use no additional data cleaning or clipping.

Table 1: Complete data-generation parameters for the headline boosted N64 Navier–Stokes dataset. Generation seed offsets separate the three independently generated splits.

Item	Value
Domain / grid	periodic $[0, 1)^2$, 64×64
Train / validation / test	1,024 / 128 / 256
Dataset seed	31; split offsets 0 / 100,000 / 200,000
Final time / time step	$T = 0.5$ / $\Delta t = 10^{-3}$
Viscosity / forcing	10^{-3} / none
Integrator	pseudo-spectral RK4, 500 steps
Aliasing control	tensor-product 2/3 filter
Initial-condition filter	$\exp(-\ k\ _2^2/8)$
IC centering / scaling	zero spatial mean; SD $+10^{-6}$; amplitude 1
Ambient velocity	$c_x, c_y \stackrel{\text{iid}}{\sim} \mathcal{U}[-0.5, 0.5]$
Generator batch / precision	8 / <code>float32</code>
Input / target shapes	$64 \times 64 \times 3$ / $64 \times 64 \times 1$
Processed SHA-256	<code>defc9703dbc41494fabf9fcfe9e3408eb2e1c38908</code> <code>14e47fe03d0f6b849e3f50</code>
Config fingerprint	<code>b2b5dec2dd3049ff491071674c34042c89f8c1a855</code> <code>f86cbb199614bfafed0cdd</code>

Splits and labeled subsets. The train, validation, and test fields are generated independently and stored before model fitting. Validation and test splits are always used in full. For a training fraction $q < 1$, a run with seed s retains the first $\max(1, \text{round}(1024q))$ indices of a `randperm` generated with seed s . The resulting 1%, 2%, 5%, and 10% subsets therefore contain 10, 20, 51, and 102 labeled examples. Methods with the same seed and fraction use the same labeled subset. Consequently, variation across training seeds includes both parameter initialization and labeled-subset variation, whereas the generated dataset itself remains fixed.

FNO2d architecture. Every headline row uses the same FNO2d with three input channels and one output channel, width 48, 12 retained modes on each axis, and four Fourier blocks. Each block sums a learned spectral convolution and a 1×1 pointwise convolution and applies GELU. A pointwise $48 \rightarrow 96 \rightarrow 1$ projection with GELU produces the output. We use no coordinate channels, spatial padding, normalization layers, or dropout. The model has 2,668,609 trainable parameters. LOCO adds no parameters or inference-time operation.

DeepONet architecture and backbone-control protocol. The Reviewer 2 backbone control uses a fixed-grid DeepONet with the same 64×64 , three-channel inputs and one-channel outputs. Its branch encoder has four circularly padded 3×3 , stride-2 convolutions with channel widths $3 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256$, GELU after every convolution, and a $4096 \rightarrow 480 \rightarrow 256$ GELU projection. We deliberately use no normalization: normalizing spatially constant channels could erase the two ambient-velocity inputs. The trunk receives 48 fixed periodic coordinate features, $\{\sin(2\pi kx), \cos(2\pi kx), \sin(2\pi ky), \cos(2\pi ky)\}_{k=1}^{12}$, and applies a $48 \rightarrow 256 \rightarrow 256 \rightarrow 256 \rightarrow 256$ MLP with GELU after the first three linear maps. A rank-256 branch–trunk inner product, divided by $\sqrt{256}$, plus one learned scalar bias produces each output-grid value. The model has 2,688,033 trainable parameters, 0.73% more than the FNO2d. The circular padding and periodic query features encode the domain topology; there is no spectral convolution and no hard-coded Galilean input–output action. The sensor and query grids are fixed at N64, so this experiment does not test resolution transfer.

We reproduce all four headline training methods with seeds $\{23, 31, 47, 59, 71\}$ and predeclare DeepONet augmentation versus DeepONet augmentation plus normalized LOCO as the primary paired comparison. It shares the dataset, seed-indexed 20-example labeled subsets, transform distribution, optimization hyperparameters, and validation rule with the FNO protocol. All four DeepONet methods use the same DeepONet inference graph and parameter count. Methods share the initialization seed and labeled subset, but their full stochastic training trajectories are not common-random-number coupled: the schedules restart the 20-example loader a different number of times, and the LOCO method consumes an additional independent

Table 2: Headline N64 training variants. At 2% labels, batches cycle through sizes 16 and 4. The transformed-example totals below use the actual partial batches rather than treating every update as a full batch.

Method	Steps/epoch	Updates	Batch forwards	λ_{orb}	Aug. examples	Orbit examples
FNO	4	600	600	–	0	0
FNO+LOCO	4	600	1,200	0.05	0	6,000
FNO+augmentation	6	900	1,800	–	9,000	0
FNO+augmentation+LOCO	4	600	1,800	0.10	6,000	6,000
DeepONet	4	600	600	–	0	0
DeepONet+LOCO	4	600	1,200	0.05	0	6,000
DeepONet+augmentation	6	900	1,800	–	9,000	0
DeepONet+augmentation+LOCO	4	600	1,800	0.10	6,000	6,000

transform draw per update. Validation and test transform draws are seed-matched across methods. We fixed the architecture and copied the FNO coefficients 0.05 (LOCO only) and 0.10 (augmentation plus LOCO) before completing the five-seed test outcomes; there was no DeepONet-specific test-set or lambda sweep.

Optimization and checkpointing. Training uses `float32` without automatic mixed precision. We optimize with AdamW for 150 epochs using learning rate 10^{-3} , weight decay 10^{-4} , $(\beta_1, \beta_2) = (0.9, 0.999)$, optimizer $\epsilon = 10^{-8}$, and no AMSGrad. A cosine-annealing schedule spans all 150 epochs, has zero minimum learning rate, and uses no warm-up. We clip the global gradient norm at 1.0. The shuffled training loader uses batch size 16, no workers, and retains the final partial batch; validation and test loaders are not shuffled. Validation runs every 10 epochs on all 128 examples. We do not early-stop and select the checkpoint with the lowest validation ID mean per-example relative L^2 .

The supervised and augmented terms are the implemented unsquared per-example relative L^2 losses in Section 3, with $\lambda_{\text{aug}} = 1$. The finite-orbit numerator is a mean square over spatial/output entries and is divided by $\epsilon^2 + \eta$, with $\eta = 10^{-6}$. On every relevant update, augmentation and LOCO draw independent transforms; within each draw, transform parameters are sampled independently for every example. Gradients propagate through both LOCO prediction branches.

The primary fixed-weight comparison uses $\lambda_{\text{orb}} = 0.10$; the orbit-only row uses 0.05. The robustness sweep evaluates $\lambda_{\text{orb}} \in \{0.01, 0.05, 0.10, 0.20, 0.30\}$, and we identify 0.30 explicitly when reporting the post-hoc tuned row. Unless a table says otherwise, five-seed results use seeds $\{23, 31, 47, 59, 71\}$, while compact controls use $\{23, 31, 47\}$.

Label-efficiency update schedules. The 1%, 5%, and 10% comparisons contain 10, 51, and 102 labeled examples, respectively, and use batch size 16 for 150 epochs. At every fraction, augmentation uses six optimizer updates per epoch (900 total), with two batch-level model evaluations per update, whereas augmentation plus normalized LOCO uses four updates per epoch (600 total), with three evaluations per update. Thus both methods use 12 batch-level model evaluations per epoch and 1,800 over training, although the numbers of optimizer updates and backward passes differ. Accounting for retained partial batches, augmentation applies 9,000, 12,450, and 14,400 per-example transforms over training at 1%, 5%, and 10%; augmentation plus LOCO applies 6,000, 7,650, and 9,600 augmentation transforms and the same respective numbers of independently drawn orbit transforms. At 5%, the fourth batch contains only three examples, so the comparison matches batch-level but not per-example model evaluations.

Galilean training and transformed-test distributions. For a relative boost $\delta = (\delta_x, \delta_y)$, the input action adds δ to the two ambient-velocity channels and the output action applies the exact periodic spectral shift $\omega_T(x) \mapsto \omega_T(x - \delta T)$. During ordinary training and test evaluation, $\delta_x, \delta_y \stackrel{\text{iid}}{\sim} \mathcal{U}[-0.25, 0.25]$. The severity study instead uses component bounds $r \in \{0.10, 0.20, 0.25, 0.35, 0.50\}$. These are additional relative boosts of the same held-out examples, not a separately simulated test split; because the original ambient frame is in $[-0.5, 0.5]^2$, the transformed observable frame at $r = 0.50$ can extend to $[-1, 1]^2$, although not every draw lies outside the original componentwise support. We therefore use “symmetry-shifted OOD” in this specific sense.

Table 3: Secondary dataset generation, curation, and preprocessing. All processed tensors are `float32`.

Dataset	Domain, splits	resolution,	Evolution parameters	Initial data or source	Preparation and processed SHA-256
1D viscous Burgers	Periodic $[0, 1]$, $N = 128$; 2,048/256/512		$T = 0.5$, $\Delta t = 10^{-3}$, $\nu = 0.01$; pseudo-spectral RK4, 500 steps, 2/3 de-aliasing; generator batch 64	Modes $k = 1, \dots, 8$, independent Gaussian sine/cosine coefficients scaled by $k^{-1.5}$; zero mean, unit SD, amplitude 0.8, mean 0; dataset seed 23 and split offsets 0/100,000/200,000	No further normalization or exclusions; shapes $128 \times 1 \rightarrow 128 \times 1$; 1753fa2e5ca4fb474f93fc8cdef5732daea94b7d36d7537113b38f9e3d029758
Unboosted N64 Navier-Stokes (D4)	Periodic $[0, 1]^2$, 64^2 ; 1,024/128/256		Same T , Δt , ν , solver, de-aliasing, and generator batch as the headline data; no ambient velocity	Same filtered Gaussian vorticity law (smoothness 8, amplitude 1); dataset seed 31 and split offsets 0/100,000/200,000	One input and one target vorticity channel; no further normalization; defade883926ad6ba6ef469bf1e6ba69b3f7d57e743e4f53c83a05acb88a8f7e
1D Dirichlet heat diagnostic	$[0, 1]$, $N = 64$ including endpoints; 512/128/128		$u_t = 0.01u_{xx}$, $T = 0.1$; exact eight-mode sine-series decay $e^{-0.01(\pi k)^2 T}$	Eight sine modes with independent Gaussian coefficients scaled by $1/k$, samplewise unit-SD normalization, amplitude 1; split seeds 23/34/45	Homogeneous Dirichlet endpoints; no further normalization; shapes $64 \times 1 \rightarrow 64 \times 1$; 7427ec21304ae8879e700653e65ee2393416b2daa8bc4b27756c1cae1ff09b2d
rMD17 ethanol [6, 7]	Fixed atom nine-molecule; 500/1,000/1,000		Static force regression; no PDE solve	Released <code>rmd17_ethanol.npz</code> ; raw SHA-256 0824d78c687b8ede25f7162c54c5c6434cfe37f9f859ee8f223c60379d7d5cf9; historical custom selection with metadata seed 2027, rather than the official split CSVs	Subtract each conformation's arithmetic coordinate centroid; append nuclear charge $Z/10$; targets are unnormalized forces. Exact cached 500-label tensor is preserved in the artifact with raw <code>old.indices</code> ; SHA-256 fb5515b3355651459c2ad9078e7cfb5bb63dce96e55e7c36956cbe9ff25e5472. Coordinates in Å, forces in kcal mol ⁻¹ Å ⁻¹

Evaluation, uncertainty, and latency. For each checkpoint, ID relative L^2 averages one per-example ratio over all 256 test examples. Orbit OOD error and relative equivariance defect average over four independent transforms per example (1,024 transformed pairs); the standard test transform RNG seed is $s + 200,000$. The severity sweep also uses four draws per example, with seed $s + 700,000$. Tables report the arithmetic mean and sample standard deviation (`ddof=1`) across seeds. Paired comparisons use matched-seed differences and two-sided 95% Student- t intervals with $n - 1$ degrees of freedom.

Latency is measured on the first 16-example test batch using five unmeasured warm-up forwards followed by 20 timed forwards; milliseconds per sample is elapsed time divided by 20×16 . The retained headline manifests record CPU execution, Windows platform string `Windows-10-10.0.26100-SP0`, Python 3.11.6, and PyTorch 2.12.0+cpu. The host currently identifies its processor as an 11th Gen Intel Core i7-1185G7 at 3.00 GHz; the original manifests did not record the CPU model, RAM, BLAS thread count, or peak memory, so latency should be interpreted as host-specific. Deterministic kernels were not forced (`deterministic=false`), but every run stores its resolved config, config hash, Git commit, dataset SHA-256, checkpoint, metrics, and initial/final Torch RNG states.

The DeepONet campaign uses the same host and software versions and records `OMP_NUM_THREADS=1` and `MKL_NUM_THREADS=1`. Its seed shards were executed concurrently as uninterrupted direct runs; checkpoint-chunk resume was not used. The recorded run wall times and latency samples are audit fields rather than controlled cross-method or cross-backbone efficiency comparisons. The unchanged-inference statement follows directly from using the identical DeepONet graph and parameter count with and without LOCO.

5.2 Secondary datasets and model hyperparameters

Table 3 records the complete preparation of the secondary benchmarks. These experiments are reported as scope and mechanism checks; the headline claim remains the boosted N64 setting above.

Burgers transform mixture. The reported Burgers comparison uses all 2,048 labeled training pairs. Its orbit metrics are not Galilean-only. For each minibatch, we choose either a periodic translation (probability 0.5, shift $s \sim \mathcal{U}[-0.25, 0.25]$) or a Galilean boost (probability 0.5, $b \sim \mathcal{U}[-0.35, 0.35]$); parameters within the selected family are then independent across examples. The Galilean action is $u_0(x) \mapsto u_0(x) + b$ and $u_T(x) \mapsto u_T(x - bT) + b$. Severity multipliers 1, 1.5, 2, 3 produce translation bounds 0.25, 0.375, 0.50, 0.75 and boost bounds 0.35, 0.525, 0.70, 1.05. For a translation draw the scalar step is $\epsilon = \max(|s|, 10^{-6})$, and for a boost draw it is $\epsilon = \max(|b|, 10^{-6})$; the normalized orbit term divides by $\epsilon^2 + \eta$ with $\eta = 10^{-6}$. Standard

Table 4: Model and optimization hyperparameters for every reported benchmark. Settings not repeated in the last column use the AdamW/cosine defaults stated in the main protocol.

Benchmark	Architecture	Parameters	Epochs / batch	Benchmark-specific training and transforms
Boosted N64	FNO2d, 3 $\rightarrow$ 1, width 48, modes 12×12 , depth 4, no grid, GELU	2,668,609	150 / 16	2% headline has 20 labels and 4 steps/epoch; augmentation compute match uses 6; $\lambda_{\text{aug}} = 1$, fixed $\lambda_{\text{orb}} = 0.10$, $\eta = 10^{-6}$
Boosted N64 / DeepONet	Circular CNN branch 3 $\rightarrow$ 32 $\rightarrow$ 64 $\rightarrow$ 128 $\rightarrow$ 256, 4096 $\rightarrow$ 480 $\rightarrow$ 256 head; 12-harmonic periodic coordinate trunk, width/rank 256	2,688,033	150 / 16	Same 20-label subsets, seeds, optimizer, transform law, and 4/4/6/4 step schedules as the FNO four-method protocol; fixed $\lambda_{\text{orb}} = 0.10$ for the primary Aug.+LOCO row
Burgers	FNO1d, 1 $\rightarrow$ 1, width 64, 16 modes, depth 4, no grid, GELU	287,361	150 / 32	Full 2,048 labels, 64 steps/epoch; seeds 23/31/47; orbit-only $\lambda = 0.05$, Aug.+LOCO $\lambda = 0.25$
Dirichlet heat	FNO1d, 1 $\rightarrow$ 1, width 32, 12 modes, depth 4, coordinate grid, GELU	55,649	80 / 32	10%=51 labels, 2 steps/epoch; seeds 23/31/47; $\lambda_{\text{aug}} = 1$, $\lambda_{\text{orb}} = 0.05$, $\eta = 10^{-6}$
rMD17 ethanol	Flatten 9×4 input; four width-128 SiLU linear layers; 27-output linear head reshaped to 9×3	57,755	150 / 32	Reported 500-label comparison: Aug. 24 and Aug.+LOCO 16 steps/epoch (48 batch forwards/epoch), seeds 23/31/47/59/71; validation every 25 epochs; $\lambda_{\text{aug}} = 1$, $\lambda_{\text{orb}} = 0.03$, $\eta = 10^{-3}$; step normalization enabled
D4 FNO	FNO2d, 1 $\rightarrow$ 1, width 48, modes 12×12 , depth 4, no grid, GELU	2,668,513	150 / 16	1%=10 labels, 4 steps/epoch; seeds 23/31/47/59/71; Aug.+LOCO $\lambda = 0.20$, $\eta = 10^{-6}$
D4 G-FNO	D4 group FNO, pseudoscalar input/output, width 8, modes 6×6 , depth 4, no grid	350,280	150 / 16	Supervised only; same 10-label subsets, 4 steps/epoch, and five seeds; architectural comparator is not parameter- or mode-matched

evaluation uses four draws for each of 512 test examples (2,048 transformed example-draw pairs); the severity analysis uses eight.

rMD17 source, split, and preprocessing. We use the fixed-order, nine-atom ethanol trajectory from the revised MD17 release [6, 7]. For each conformation, we subtract the arithmetic (not mass-weighted) coordinate centroid and append the scaled nuclear charge to each atom:

$$X_c = X - \mathbf{1}\bar{x}^\top, \quad a_i = (X_{c,i}, Z_i/10), \quad u_i = F_i.$$

Coordinates are in Å and force targets in kcal mol $^{-1}$ Å $^{-1}$. Inputs and targets are stored as `float32`; forces are otherwise unnormalized, and energy labels are not used. All optimization seeds {23, 31, 47, 59, 71} reuse the same mutually disjoint 500/1,000/1,000 training/validation/test split.

This historical custom selection does not use the release’s official split CSVs. Its metadata records split seed 2027, but the historical sampler was not retained and the current seeded `torch.randperm` does not reconstruct the selection. The exact operational split rule is therefore to look up each split’s stored `old.indices` in the raw NPZ’s unique `old.indices` array and preserve the cached order. The preserved tensor and its SHA-256 in Table 3 are the authoritative evidence input. The released verifier maps all 2,500 identifiers back to the hashed NPZ with zero `float32` input and force discrepancy. The older 1,000-label cache has different validation/test slices and is not used as a controlled label-scaling comparison.

rMD17 rigid motions, losses, and evaluation. Each per-example transform draws $z \sim \mathcal{N}(0, I_3)$, sets $q = z / \max(\|z\|_2, 10^{-8})$, draws $\theta \sim \mathcal{U}[-\pi/2, \pi/2]$, and draws translation components $t_j \stackrel{\text{iid}}{\sim} \mathcal{U}[-1, 1]$ Å. If $Q(q, \theta)$ is the axis-angle rotation, then

$$T_g^A(X_c, Z) = (X_c Q^\top + \mathbf{1}t^\top, Z), \quad T_g^U F = F Q^\top.$$

Coordinates are not recentered after this transformation. The axis is uniform on the sphere and the bounded angle is uniform on its interval; this is not a Haar-uniform rotation draw. The implemented composite step is $\epsilon = \max(|\theta| + \|t\|_2, 10^{-6})$. Because it combines radians and Å, it is a fixed disclosed weighting rather than a canonical, unit-free SE(3) norm.

The supervised and augmented terms use the minibatch mean of the unsquared per-conformation relative force error,

$$\ell_{\text{force}}(\hat{F}, F) = \frac{\|\hat{F} - F\|_F}{\max(\|F\|_F, 10^{-8})},$$

not force MAE or force MSE. The molecular orbit term is the per-conformation mean squared force-entry defect divided by $\epsilon^2 + 10^{-3}$. The reported Aug.+LOCO row uses $\lambda_{\text{aug}} = 1$ and $\lambda_{\text{orb}} = 0.03$; augmentation and LOCO draw independent rigid motions, and gradients propagate through both prediction branches. Force MAE is an evaluation metric only.

For the reported 500-label comparison, augmentation performs 24 updates per epoch with two model evaluations per update, whereas Aug.+LOCO performs 16 updates with three evaluations per update. Both therefore use 48 batch-level model evaluations per epoch and 7,200 over 150 epochs. Because 500 is not divisible by 32, this is not exact at the example level: augmentation visits 756 examples and samples 756 rigid motions per epoch (113,400 over training), giving 1,512 sample-level model evaluations per epoch, whereas Aug.+LOCO visits all 500 examples once and independently samples one augmentation and one orbit motion for each (1,000 per epoch; 150,000 total), giving 1,500 sample-level model evaluations per epoch. Validation occurs every 25 epochs, and the checkpoint is selected by untransformed validation relative force L^2 . Final evaluation averages four independent motions for each of the 1,000 test conformations (4,000 conformation-motion pairs per run), with transform RNG seed $s + 200,000$. The evaluation ranges equal the training ranges, so these are described as transformed-input rather than wider-severity OOD metrics. We use 10 latency warm-ups and 50 repeats.

Finite D_4 action. For the D_4 experiment, a rotation index is uniform on $\{0, 1, 2, 3\}$ and reflection is Bernoulli(0.5), giving all eight elements uniformly. Vorticity is treated as a pseudoscalar and changes sign under reflection. We set $\epsilon = 1$, so this is finite-group consistency rather than a local Lie-algebra step. Augmentation and Aug.+LOCO both use four updates per epoch; their forward-pass compute is therefore not matched. Evaluation is Monte Carlo rather than an exhaustive average over all eight elements: for each of the 256 held-out examples, it samples four independent uniform D_4 elements with replacement, so repetitions and the identity are possible, and averages the resulting 1,024 example-element pairs. For optimization seed s , the evaluation RNG is seeded by $s + 200,000$, which matches the draws across methods with the same seed.

Boundary diagnostic. The heat-equation shift is sampled as $s \sim \mathcal{U}[-0.1, 0.1]$ and implemented by zero-padded linear interpolation. The masked variant evaluates only source coordinates in $[0.05, 0.95]$. Because a fixed Dirichlet problem is not globally translation-equivariant and boundary influence propagates into the interior, this experiment is an approximate boundary-mask diagnostic, not an exact-symmetry OOD benchmark. Its transformed target is the shifted cached output, not a newly solved heat equation for the shifted input. To avoid comparing checkpoints on different scoring supports, we re-evaluate all three training variants with the same transform draws under both a full-domain score and a common valid-overlap score with margin 0.05. Masking restricts where this approximate constraint is scored; it does not restore translation equivariance of the fixed-boundary heat equation. Here $\epsilon = \max(|s|, 10^{-6})$ and $\eta = 10^{-6}$. The common-support re-evaluation draws four shifts for each of 128 test examples, hence 512 transformed example-draw pairs per checkpoint under each scoring support. It uses evaluation seed $s_{\text{run}} + 700,000$; resetting that seed for every evaluation matches the draws across all three methods and both scoring supports.

5.3 Additional N64 controls

Step-size and tangent controls. At 2% labels, the denominator sensitivity uses $\eta \in \{10^{-8}, 10^{-6}, 10^{-4}\}$ at component radius 0.25, while the step-size sensitivity uses radii $\{0.125, 0.25, 0.50\}$ at $\eta = 10^{-6}$; each uses seeds 23, 31, and 47. The tangent-propagation comparison uses the same 20 labels, four steps per epoch, all five headline seeds, and $\lambda_{\text{tangent}} = 0.10$. Its input tangent changes only the two ambient velocity channels, and its output tangent is $-T(\delta_x \partial_x^{\text{spec}} + \delta_y \partial_y^{\text{spec}}) \mathcal{G}(a)$. These are Fourier spectral derivatives, obtained by differentiating the same Fourier phase-shift action used for the finite Galilean transform. For $u \in \mathbb{R}^{H \times W}$ on

$[0, L)^2$, the exact discrete implementation is

$$\hat{u} = \text{FFT}_2(u), \quad \partial_x^{\text{spec}} u = \Re \text{IFFT}_2[(2\pi i f_x) \hat{u}], \quad \partial_y^{\text{spec}} u = \Re \text{IFFT}_2[(2\pi i f_y) \hat{u}],$$

where $f_x = \text{fftfreq}(W, L/W)$ and $f_y = \text{fftfreq}(H, L/H)$ in standard FFT ordering and are broadcast along their respective axes. At $N = 64, L = 1$, each frequency vector is $(0, 1, \dots, 31, -32, -31, \dots, -1)$. The tangent calculation applies no finite-difference stencil, Fourier-mode truncation, filtering, or de-aliasing. A differentiable JVP is computed with graph construction enabled, and the tangent mean square is divided by $\|\delta\|_2^2 + 10^{-6}$. Augmentation+tangent draws its supervised augmentation independently.

The plain finite-transform control uses the same data, architecture, transform range, augmentation, four-update schedule, and five headline seeds $\{23, 31, 47, 59, 71\}$ as normalized LOCO, but minimizes the raw finite equivariance MSE. Since $\delta_x, \delta_y \stackrel{\text{iid}}{\sim} \mathcal{U}[-0.25, 0.25]$, $\mathbb{E}\|\delta\|_2^2 = 2(0.25)^2/3 = 1/24$. We therefore set its raw-loss weight to 2.4, an approximate leading-order average-scale match to the normalized $\lambda_{\text{orb}} = 0.10$ objective. The square sampling law couples direction and norm, so this is not an exact equivalence, nor an assertion that the two nonlinear losses have identical realized magnitudes.

Semi-supervised joint training. The joint 2%-label experiment uses supervised and augmented batches from the 20-example labeled subset and separate LOCO batches from the full 1,024-example training pool. It performs four combined updates per epoch, uses full-pool orbit batch size 16, and therefore samples 64 (not all 1,024) full-pool orbit examples per epoch; $\lambda_{\text{orb}} = 0.05$ and seeds are 23, 31, and 47.

Unlabeled target adaptation. The target-adaptation table is transductive: the same 256 test inputs are used without labels during adaptation and with labels for before/after evaluation. Starting from the 2%-label augmentation or fixed-weight Aug.+LOCO checkpoint, we update only the FNO projection layers (4,801 parameters) for five epochs and 80 updates. We use AdamW with learning rate 10^{-5} , zero weight decay, no scheduler, batch size 16, and gradient clip 0.1. The objective combines orbit loss with a unit-weight relative prediction-preservation term; parameter anchoring has weight zero. Target and orbit boosts are independently redrawn at component radius 0.35 or 0.50. Evaluation uses four transforms per example and seeds 23, 31, and 47.

Canonicalization diagnostics. The metric called oracle paired-source error compares $T_g \mathcal{G}(a)$ with $T_g u$; it does not evaluate $\mathcal{G}(T_g a)$ and is therefore a diagnostic rather than a new model. Observable canonicalization is also evaluation-only: it reads the total ambient velocity from the input, sets those channels to zero, evaluates the FNO once, and shifts the result by cT . It changes preprocessing but neither retrains the checkpoint nor changes the number of model calls.

6 Experiments

Question. We test whether local orbit consistency reduces symmetry-induced out-of-distribution error and equivariance defect for an unchanged neural-operator backbone. The main experiment uses a periodic 64×64 2D Navier–Stokes vorticity operator with Galilean boosts. The base model is the same FNO2d architecture in every headline row; orbit consistency changes training only and does not add an inference-time module. A near-capacity-matched DeepONet reproduction on the same data tests whether the observed effect requires Fourier layers.

Protocol. We compare four training methods: supervision, supervised Lie augmentation, supervised orbit consistency, and augmentation plus orbit consistency. The headline low-label setting uses 2% labeled data and five seeds. Within both the FNO and DeepONet studies, the forward-evaluation-matched augmentation baseline uses six update steps per epoch, while augmentation plus orbit consistency uses four update steps per epoch with $\lambda_{\text{orb}} = 0.10$. Thus the primary pair uses 12 batch-level operator evaluations per epoch, but not the same optimizer-update or backward-pass count. Every within-backbone row has an unchanged test-time architecture. The FNO headline artifacts are regenerated by `scripts/reproduce_paper_artifacts.py`; reviewer-specific controls, including DeepONet, are regenerated from their recorded runs by `scripts/run_reviewer_followup_experiments.ps1`.

Table 5: N64 Galilean headline at 2% labels over five seeds. All rows use the same FNO2d architecture with 2,668,609 parameters. Lower is better for relative L^2 , orbit OOD relative L^2 , and equivariance defect.

Method	Steps/epoch	Relative L^2	Orbit OOD L^2	Eq. defect	Latency ms/sample
FNO	4	0.9287 ± 0.0228	0.9568 ± 0.0223	0.6988 ± 0.0173	16.53 ± 2.13
FNO + orbit	4	0.4646 ± 0.0394	0.4952 ± 0.0348	0.3359 ± 0.0170	16.11 ± 1.66
FNO + aug.	6	0.5009 ± 0.0590	0.5276 ± 0.0525	0.3816 ± 0.0322	18.76 ± 7.34
FNO + aug. + orbit	4	0.3676 ± 0.0360	0.3950 ± 0.0334	0.2677 ± 0.0174	16.87 ± 3.19

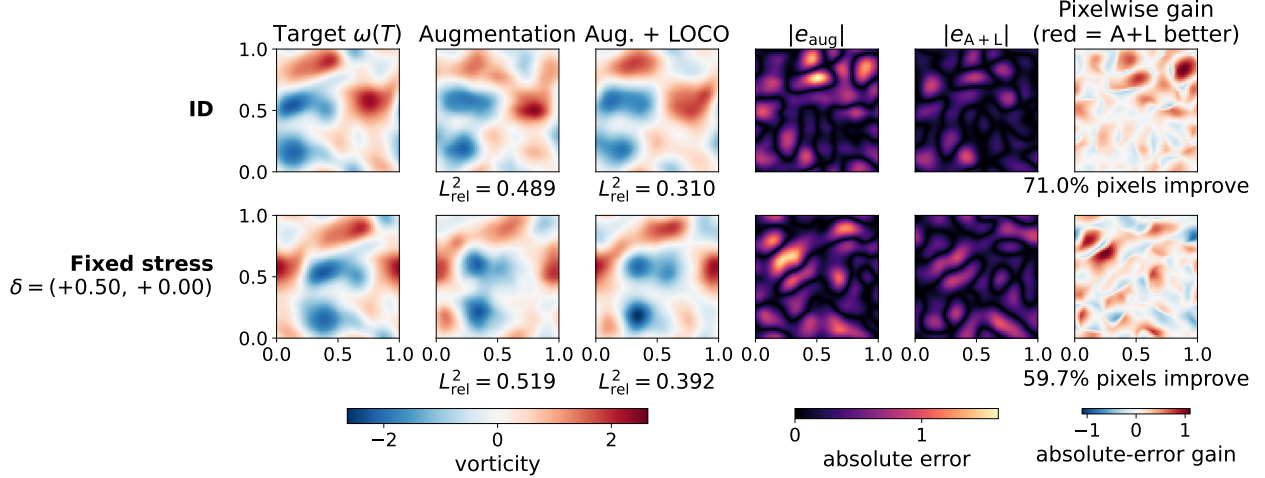


Figure 4: Fields for one held-out N64 Navier-Stokes case: the upper row is in distribution (ID), and the lower row is a deterministic fixed Galilean stress $\delta = (0.50, 0)$, twice the componentwise training radius. The 2%-label checkpoints use training seed 23; the displayed case is zero-based test index 97, chosen after evaluation by the outcome-conditioned rule “paired fixed-stress relative- L^2 improvement closest to the median” over all 256 test cases. The stress changes the ambient frame from $(0.073, -0.123)$ to $(0.573, -0.123)$ without changing the initial vorticity. Its target is constructed from the cached held-out target by the exact periodic Galilean output action (with solver closure validated separately). Columns show the target, direct predictions from forward-evaluation-matched augmentation and augmentation plus LOCO (A+L; $\lambda_{\text{orb}} = 0.10$), their pointwise absolute errors, and the gain $|e_{\text{aug}}| - |e_{\text{A+L}}|$; positive (red) gain means smaller A+L error. Field and absolute-error panels share their respective color ranges across rows.

Main N64 Galilean result. Table 5 shows that at 2% labels, augmentation plus orbit consistency improves both paper-facing symmetry metrics over forward-evaluation-matched supervised augmentation. Orbit OOD relative L^2 decreases from 0.5276 ± 0.0525 to 0.3950 ± 0.0334 , a 25.1% reduction. Equivariance defect decreases from 0.3816 ± 0.0322 to 0.2677 ± 0.0174 , a 29.8% reduction. Relative to the plain supervised FNO baseline, the reductions are 58.7% for orbit OOD error and 61.7% for equivariance defect. Parameter counts are identical across the headline rows.

Illustrative fixed-stress fields. Figure 4 complements the aggregate metrics with direct field predictions. For the disclosed, outcome-conditioned seed-23 example, augmentation plus LOCO lowers relative L^2 from 0.489 to 0.310 in distribution and from 0.519 to 0.392 under the fixed axis-aligned stress. Its absolute error is lower on 59.7% of fixed-stress pixels, while local negative regions in the gain map show that the improvement is not spatially uniform. The displaced residual structure is consistent with the periodic action; the transformed target itself is obtained by applying the exact periodic output action to the cached held-out target, rather than treating a domain edge as a boundary.

Full-test selection and gradient context. Figure 5 places that selected field among all 256 held-out cases evaluated with the same seed-23 fixed stress. Augmentation plus LOCO has lower fixed-stress relative

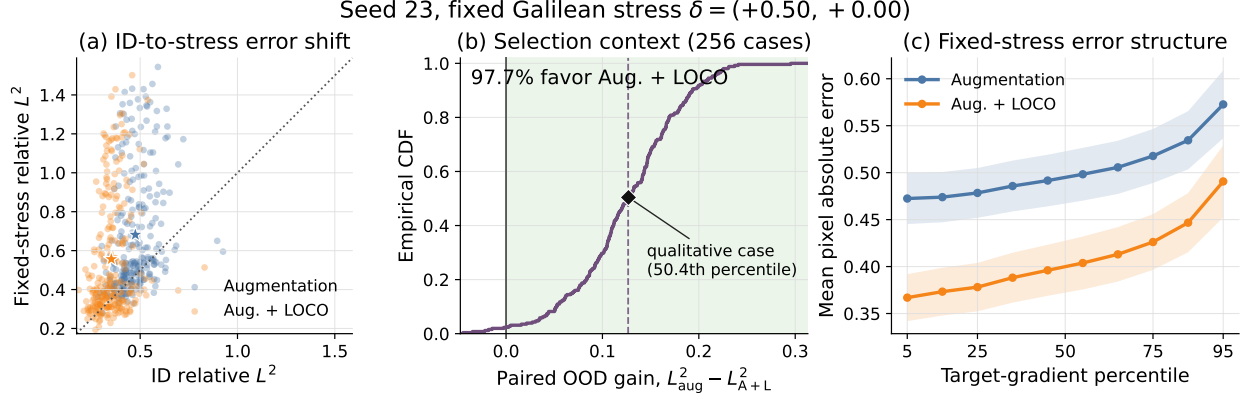


Figure 5: Full-test context for the seed-23, 2%-label fixed Galilean stress $\delta = (0.50, 0)$. (a) Each point is one of the 256 held-out cases, plotted by ID and fixed-stress relative L^2 ; the dotted line is equality and stars mark method means. (b) Empirical CDF $\hat{F}_{\Delta}(x) = 256^{-1} \sum_{i=1}^{256} \mathbf{1}\{\Delta_i \leq x\}$ of the per-example paired gain $\Delta_i = e_{\text{aug},i}^{\delta} - e_{\text{A+L},i}^{\delta}$, where e^{δ} is fixed-stress relative L^2 ; positive values favor augmentation plus LOCO. The diamond locates the outcome-conditioned qualitative case in Figure 4. Its zero-based test index is 97; its gain of 0.1270 is closest to the sample median and lies at the 50.4th empirical percentile ($\hat{F}_{\Delta} = 129/256 = 0.5039$). Overall, 250/256 cases favor A+L and the mean gain is 0.1253. (c) Mean pixel absolute error in equal-mass bins of fixed-stress target-gradient magnitude, aggregated first within each example and then across examples; shading gives pointwise normal-approximation 95% intervals across the 256 per-example bin means. Panel (c) is a descriptive gradient-conditioned diagnostic, not an independent inferential test.

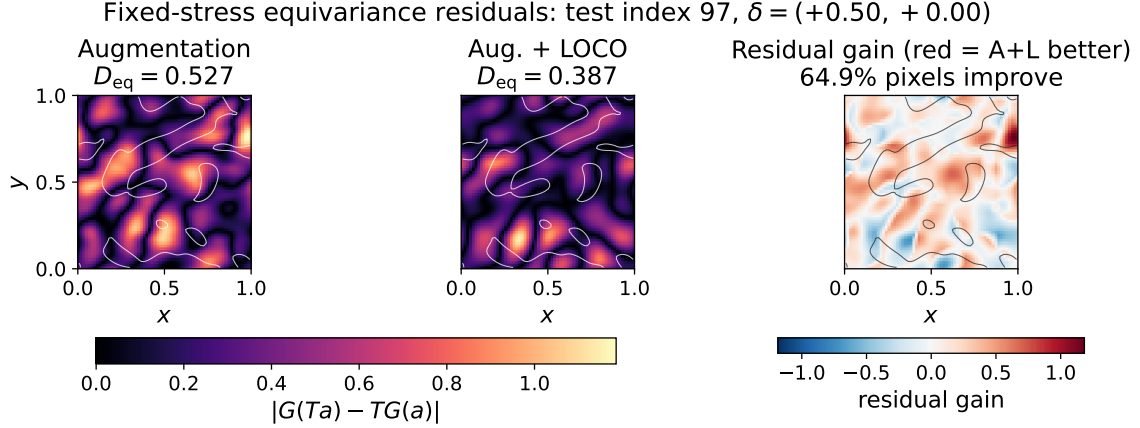


Figure 6: Spatial equivariance residuals for the same seed-23, 2%-label, zero-based test index 97 and fixed stress $\delta = (0.50, 0)$. The left and center panels show $|G(Ta) - TG(a)|$ for augmentation and augmentation plus LOCO on a common color scale; their relative equivariance defects are 0.527 and 0.387. The right panel shows residual gain, $|G_{\text{aug}}(Ta) - TG_{\text{aug}}(a)| - |G_{\text{A+L}}(Ta) - TG_{\text{A+L}}(a)|$, on a symmetric scale, with positive (red) values favoring A+L; A+L has lower absolute residual on 64.9% of pixels. Contours mark the top quartile of fixed-stress target-gradient magnitude (white in the residual panels and dark in the gain panel) to provide spatial context, not to establish a causal association.

L^2 in 250/256 cases, with mean paired gain 0.1253; the selected zero-based test index 97 lies at the 50.4th empirical percentile ($\hat{F}_{\Delta} = 129/256 = 0.5039$), documenting that it is near the median of this outcome-conditioned selection statistic. The corresponding full-test mean errors are 0.682 for augmentation and 0.556 for augmentation plus LOCO. When absolute error is stratified into equal-mass bins of target-gradient magnitude, both methods tend to incur more error in higher-gradient bins and augmentation plus LOCO is

Table 6: Training compute accounting for the headline 2% N64 Galilean runs. Forward/backward counts are computed from logged optimizer steps. Transformed-sample counts use the actual alternating minibatch sizes 16 and 4 for the 20-example labeled subset, including iterator restarts; wall-clock minutes are estimated from immutable RNG checkpoint timestamps for these legacy cached runs, and new runs now write `train_wall_seconds` directly. Relative train time is normalized to the supervised augmentation baseline; batch-level forward counts are matched but backward-pass counts differ.

Method	Opt. steps	Model fwds.	Bwd. passes	Aug. labeled samples	Orbit samples	Train min.	Rel. train time	Latency ms/sample
FNO	600	600	600	0	0	14.6 ± 0.4	$0.71\times$	16.53 ± 2.13
FNO + orbit	600	1200	600	0	6000	18.7 ± 0.6	$0.91\times$	16.11 ± 1.66
FNO + aug.	900	1800	900	9000	0	20.6 ± 2.5	$1.00\times$	18.76 ± 7.34
FNO + aug. + orbit	600	1800	600	6000	6000	21.0 ± 2.6	$1.02\times$	16.87 ± 3.19

Table 7: Low-label comparison for forward-evaluation-matched augmentation versus augmentation plus orbit consistency with $\lambda_{\text{orb}} = 0.10$.

Labels	Method	Orbit OOD L^2	Eq. defect	Seeds
1%	Aug.	0.7421 ± 0.0838	0.5080 ± 0.0456	3
1%	Aug. + orbit	0.6406 ± 0.0508	0.3999 ± 0.0197	3
2%	Aug.	0.5276 ± 0.0525	0.3816 ± 0.0322	5
2%	Aug. + orbit	0.3950 ± 0.0334	0.2677 ± 0.0174	5
5%	Aug.	0.3237 ± 0.0013	0.2427 ± 0.0014	3
5%	Aug. + orbit	0.2515 ± 0.0112	0.1621 ± 0.0078	3

lower in every displayed bin. This gradient-conditioned view is descriptive: its pixels and bins are spatially dependent, and it is not a separate test of why the methods differ.

Illustrative equivariance residuals. Figure 6 shows the equivariance residual for the same selected example. The relative defect decreases from 0.527 for augmentation to 0.387 for augmentation plus LOCO, which has the lower absolute residual on 64.9% of pixels. The target-gradient contours localize structure but do not establish that gradients cause the residual reduction. These three fixed-axis, seed-23 diagnostics are illustrative and outcome-aware; the randomized-severity analysis over five seeds in Figure 9 is the inferential basis for the distribution-shift claim.

Compute accounting. Table 6 reports training compute for the same headline runs. The augmentation baseline uses 900 optimizer steps and 1800 estimated model forward passes, while augmentation plus orbit consistency uses 600 optimizer steps and the same 1800 estimated model forward passes because each step evaluates the original, augmented, and orbit-transformed inputs. Thus the rows are matched in batch-level forward evaluations and have similar observed wall time; their backward-pass counts differ. Inference latency remains within run-to-run noise because the deployed map is still the same FNO2d backbone.

Low-label behavior. Figure 7 and Table 7 show lower mean errors at each tested low-label fraction. Against forward-evaluation-matched augmentation, $\lambda_{\text{orb}} = 0.10$ reduces orbit OOD error by 13.7%, 25.1%, and 22.3% at 1%, 2%, and 5% labels respectively. The corresponding equivariance-defect reductions are 21.3%, 29.8%, and 33.2%. These finite settings support a low-label benefit but do not establish a general sample-complexity law.

Backbone portability and dependence. Table 8 reproduces the full four-method experiment with a fixed-grid branch-trunk DeepONet whose parameter count differs from FNO2d by 0.73%. Relative to forward-evaluation-matched DeepONet augmentation, adding normalized LOCO lowers ID error from 0.9553 ± 0.0335 to 0.8862 ± 0.0362 (7.2%), orbit OOD error from 0.9652 ± 0.0308 to 0.8972 ± 0.0340 (7.0%), and equivariance defect from 0.6988 ± 0.0219 to 0.4921 ± 0.0295 (29.6%). The matched-seed LOCO-minus-augmentation differences are -0.06911 for ID error (95% CI $[-0.08038, -0.05784]$), -0.06799 for orbit OOD error ($[-0.08087, -0.05510]$), and -0.20674 for defect ($[-0.25557, -0.15791]$). All five seed-level differences

Table 8: DeepONet reproduction on the 2%-label N64 Galilean task over the same training seeds as the FNO comparison, {23, 31, 47, 59, 71}. All rows use the same 2,688,033-parameter architecture. The augmentation and augmentation-plus-normalized-LOCO rows are matched at 12 batch-level model evaluations per epoch; optimizer and backward-pass counts differ and are shown. Entries are mean $\pm$ sample standard deviation. Latency and wall time remain available in the per-run manifest; seed shards ran concurrently, so they are not used as efficiency comparisons.

Method	Steps/epoch	Model fws., total	Bwd. passes, total	ID L^2	Orbit OOD L^2	Eq. defect
DeepONet	4	600	600	0.9949 ± 0.0081	1.0013 ± 0.0072	0.5821 ± 0.0429
DeepONet + normalized LOCO	4	1200	600	0.9150 ± 0.0318	0.9236 ± 0.0300	0.5545 ± 0.0229
DeepONet + aug.	6	1800	900	0.9553 ± 0.0335	0.9652 ± 0.0308	0.6988 ± 0.0219
DeepONet + aug. + normalized LOCO	4	1800	600	0.8862 ± 0.0362	0.8972 ± 0.0340	0.4921 ± 0.0295

Table 9: Backbone scope over five seeds. Panel A directly applies LOCO to a non-Fourier, non-equivariant fixed-molecule MLP; ID and transformed errors are force MAE, and both rows use 48 batch-level model evaluations per epoch. The LOCO row uses the disclosed composite rigid-motion step normalization. Panel B uses relative L^2 error and contrasts an FNO training loss with a supervised hard-equivariant D4 G-FNO. The G-FNO is a saturation reference, not a LOCO-trained row, and is not parameter- or mode-matched. Metrics must not be compared across panels.

Task / backbone	Training	ID error	Transformed error	Eq. defect	Parameters	Seeds
<i>Panel A: rMD17 ethanol force prediction, 500 labels</i>						
MLP	augmentation	19.9135 ± 0.0895	20.0535 ± 0.0812	0.7372 ± 0.0139	57,755	5
MLP	augmentation + normalized LOCO	19.3686 ± 0.1389	19.4698 ± 0.1342	0.2608 ± 0.0044	57,755	5
<i>Panel B: unboosted N64 Navier-Stokes under D4</i>						
FNO2d	augmentation	0.06542 ± 0.00369	0.06538 ± 0.00366	0.04570 ± 0.00397	2,668,513	5
FNO2d	augmentation + LOCO	0.06462 ± 0.00331	0.06459 ± 0.00330	0.04499 ± 0.00383	2,668,513	5
D4 G-FNO	supervised	0.09283 ± 0.00537	0.09283 ± 0.00537	$(7.88 \pm 0.32) \times 10^{-8}$	350,280	5

have the favorable sign for all three metrics. Thus the benefit transfers to the tested second neural-operator family without changing its inference graph. The predictive reductions are nevertheless smaller than the corresponding FNO reductions (7.2% versus 26.6% for ID and 7.0% versus 25.1% for OOD), while the relative defect reductions are similar (29.6% versus 29.8%). These are descriptive within-task comparisons, not a cross-backbone significance test, and the tested DeepONet is fixed-grid and less accurate overall.

Table 9 gives two secondary reference points. In a five-seed, forward-evaluation-matched rMD17 check, adding normalized LOCO to a fixed-molecule MLP reduces force MAE by 2.7%, transformed-input force MAE by 2.9%, and force equivariance defect by 64.6%; the paired 95% intervals for all three deltas exclude zero. This is a cross-domain mechanism check, not a neural operator or an OOD-severity result. Conversely, a supervised D4 G-FNO already has numerical defect below 10^{-7} , leaving essentially no same-group consistency signal; its predictive error is higher in the reported, non-capacity-matched configuration. We therefore claim architecture-independent applicability of the objective, not architecture-independent effect size or uniform gains across neural-operator families.

Solver closure and supporting Burgers evidence. Explicit re-solves of 32 transformed test examples verify the coded Galilean identity to numerical precision: the maximum relative closure residual is 6.22×10^{-7} for N64 Navier-Stokes through component radius 0.50 and 7.51×10^{-7} for N128 Burgers through boost radius 1.05. On the learned Burgers operator, Table 10 shows that augmentation already nearly saturates the symmetry. Adding LOCO changes mean orbit-shift error from 0.00405 to 0.00381, but the paired interval crosses zero; the defect changes from 0.00265 to 0.00228, with its paired interval below zero, while mean ID error worsens from 0.00251 to 0.00265. We therefore use Burgers as an exact-action supporting check, not a second headline predictive result or a uniform accuracy improvement.

Finite versus tangent and raw consistency. Table 11 compares the finite LOCO term with an explicit JVP tangent penalty and a plain finite-transform consistency loss. On five paired seeds, normalized finite LOCO minus augmentation plus tangent is -0.04734 in ID error (95% CI $[-0.05494, -0.03974]$), -0.05397 in orbit-shift error ($[-0.06037, -0.04758]$), and -0.04799 in equivariance defect ($[-0.05209, -0.04390]$). The

Table 10: Supporting 1D Burgers exact-symmetry check over three matched seeds. The orbit-shift error difference has a paired interval crossing zero, while the equivariance-defect reduction does not.

Training	ID L^2	Orbit-shift L^2	Eq. defect
Augmentation	0.002512 ± 0.000146	0.004049 ± 0.000326	0.002653 ± 0.000326
Augmentation + LOCO	0.002655 ± 0.000003	0.003807 ± 0.000153	0.002283 ± 0.000215

Table 11: Finite-objective comparison on the N64 Galilean task at 2% labels. Entries are mean $\pm$ sample standard deviation within each panel. Both panels use five matched seeds. The augmentation baseline and both finite objectives use 12 batch-level model-forward evaluations per epoch. The tangent objective uses an explicit JVP and is not compute-matched. The raw finite weight 2.4 is an approximate leading-order average-scale match to normalized finite weight 0.10, not a tuned equivalence.

Method	Weight	Step normalization	JVP	ID L^2	Orbit-shift L^2	Eq. defect	Seeds
<i>Panel A: finite versus tangent, five matched seeds</i>							
Augmentation	$\lambda_{\text{aug}} = 1$	n/a	no	0.5009 ± 0.0590	0.5276 ± 0.0525	0.3816 ± 0.0322	5
Augmentation + tangent	$\lambda_{\text{tangent}} = 0.1$	n/a	yes	0.4150 ± 0.0363	0.4490 ± 0.0336	0.3157 ± 0.0203	5
Augmentation + normalized finite	$\lambda_{\text{orb}} = 0.1$	$\epsilon^2 + \eta$	no	0.3676 ± 0.0360	0.3950 ± 0.0334	0.2677 ± 0.0174	5
<i>Panel B: normalized versus raw finite, five matched seeds</i>							
Augmentation + normalized finite	$\lambda_{\text{orb}} = 0.1$	$\epsilon^2 + \eta$	no	0.3676 ± 0.0360	0.3950 ± 0.0334	0.2677 ± 0.0174	5
Augmentation + raw finite	$\lambda_{\text{raw}} = 2.4$	none	no	0.3779 ± 0.0421	0.4031 ± 0.0385	0.2687 ± 0.0219	5

tested finite objective therefore achieves lower means, consistent with useful information away from the identity. It does not causally isolate higher-order orbit information, however: the explicit JVP has a different computational profile and only one tangent weight was tested. The comparison therefore does not establish universal superiority over tangent propagation. On five paired seeds shared with the approximately scale-matched raw finite control, normalization changes ID error by -0.01025 (95% CI $[-0.01948, -0.00103]$), orbit-shift error by -0.00813 ($[-0.01604, -0.00021]$), and defect by -0.00103 ($[-0.00748, 0.00543]$). Thus, at this fixed radius and approximate scale match, normalization gives modest resolved reductions of 2.7% in ID error and 2.0% in orbit-shift error, while the defect difference remains unresolved. This single setting does not establish universal superiority of normalization over raw finite consistency. Its specific contribution is to remove the leading quadratic coupling between objective scale and sampled step radius.

OOD severity. Figure 9 and Table 16 evaluate Galilean boosts from below the training radius to twice the training radius. The augmentation-plus-orbit model maintains a lower orbit OOD error at every tested boost. The relative OOD-error reductions are 26.4%, 25.8%, 25.3%, 23.6%, and 20.3% at maximum boosts 0.10, 0.20, 0.25, 0.35, and 0.50. Equivariance-defect reductions follow the same trend, from 33.6% at boost 0.10 to 23.6% at boost 0.50.

Canonicalization calibration. Table 13 reports a privileged relative-boost calibration, not a deployable analytic canonicalization baseline. In the OOD severity protocol, each test input a is paired with a sampled relative boost δ , producing $T_\delta a$. The calibration uses this sampled δ and the paired in-distribution prediction $\mathcal{G}_\theta(a)$, then reports $T_\delta \mathcal{G}_\theta(a)$. This is useful for measuring how much of the direct OOD error comes from motion along the tested orbit, but it assumes access to the paired source input and sampled relative transform. It should therefore not be confused with the observable-frame analytic baseline, which reads the absolute ambient velocity c from the input channels and canonicalizes a single test input. At maximum boost 0.50, augmentation plus orbit consistency has direct OOD error 0.4910 ± 0.0216 , improving substantially over augmentation 0.6158 ± 0.0384 while remaining above its privileged calibration value 0.3676 ± 0.0360 .

Association between defect and OOD error. Figure 8 and Table 14 report a diagnostic association rather than a causal or out-of-sample prediction result. Across 90 cached N64 Galilean observations, defect correlates strongly with orbit OOD relative L^2 (Pearson $r = 0.851$, Spearman $\rho = 0.838$); after residualizing source, method, label fraction, maximum boost, and λ_{orb} , the partial Pearson correlation remains $r = 0.779$. The association also appears within the OOD severity curve alone ($r = 0.761$). Because all observations

Table 12: Fixed-boundary heat-equation diagnostic over three seeds. Every checkpoint is re-evaluated using identical transform draws and the same scoring support. “Approximate-action pseudo-target” compares with a shifted cached target, not a newly solved transformed problem. The overlap columns use the same transform-dependent source-overlap mask for every method, with source margin 0.05. Lower is better.

Training	ID L^2	Approx.-action pseudo-target, full	Defect, full	Approx.-action pseudo-target, overlap	Defect, overlap
Baseline	0.0270 ± 0.0012	0.0451 ± 0.0007	0.0341 ± 0.0008	0.0332 ± 0.0015	0.0243 ± 0.0006
Aug.+LOCO, unmasked	0.0347 ± 0.0019	0.0433 ± 0.0014	0.0246 ± 0.0008	0.0336 ± 0.0016	0.0149 ± 0.0006
Aug.+LOCO, masked	0.0363 ± 0.0012	0.0447 ± 0.0008	0.0301 ± 0.0006	0.0230 ± 0.0014	0.0117 ± 0.0004

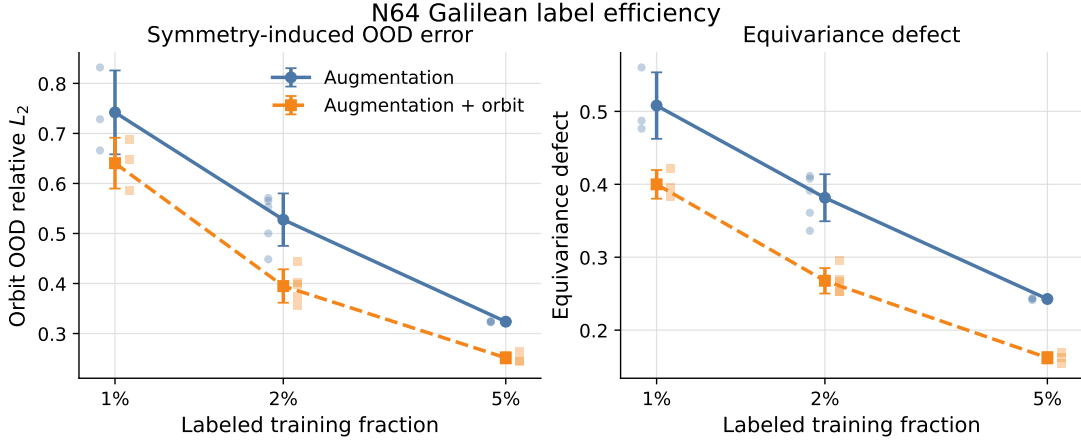


Figure 7: N64 Galilean label efficiency. Error bars show standard deviation across seeds and faint points show individual seeds. Augmentation plus orbit consistency improves orbit OOD error and equivariance defect at 1%, 2%, and 5% labels.

share datasets and pipelines, these results show that orbit consistency and OOD behavior track one another in this benchmark, not that defect is a generally validated predictor.

Boundary-condition diagnostic. Table 12 highlights a limitation rather than extending the exact-symmetry claim. Fixed Dirichlet walls break global translation equivariance, so all checkpoints are re-evaluated using the same shift draws and the same full-domain or interior support. Masked Aug.+LOCO gives the lowest valid-overlap pseudo-target error and defect, but its full-domain pseudo-target score is worse than unmasked Aug.+LOCO, and both regularized variants have worse genuine ID error than the baseline. The mask therefore localizes where an approximate constraint is useful; it does not turn the fixed-wall problem into a translation symmetry. This diagnostic motivates checking closure of the full equation, geometry, boundary data, and observation map before applying LOCO.

Ablations and scope. Table 15 reports the orbit-weight ablation at 2% labels. We keep $\lambda_{\text{orb}} = 0.10$ as the broader-experiment setting, while the completed high-weight sweep shows $\lambda_{\text{orb}} = 0.30$ as the strongest tested value. Semi-supervised joint orbit training is useful as an appendix result because it reduces equivariance defect, but it is not the strongest predictive-error result. We do not include a 3D PDE in the main study: a same-resolution 3D analogue would multiply raw field size by 64 and introduce a new solver and validation burden. The current contribution is deliberately narrow: exact-symmetry validation plus a reproducible 2D Galilean low-label result.

7 Discussion and Limitations

The evidence supports a narrow conclusion: on the solver-validated periodic N64 Galilean task, LOCO improves the tested FNO’s symmetry-shifted error and equivariance defect, with the largest observed gains

Table 13: Privileged relative-boost calibration at 2% labels. Direct OOD evaluates the model on the boosted input. The privileged calibration uses the sampled relative boost from the evaluation protocol and the paired in-distribution prediction, so it is not an observable-frame analytic baseline. Lower is better.

Max boost	Aug. direct OOD	Aug. priv. calib.	Aug. + orbit direct OOD	Aug. + orbit priv. calib.
0.10	0.5050 ± 0.0574	0.5009 ± 0.0590	0.3715 ± 0.0349	0.3676 ± 0.0360
0.20	0.5176 ± 0.0543	0.5009 ± 0.0590	0.3840 ± 0.0331	0.3676 ± 0.0360
0.25	0.5277 ± 0.0523	0.5009 ± 0.0590	0.3943 ± 0.0318	0.3676 ± 0.0360
0.35	0.5555 ± 0.0476	0.5009 ± 0.0590	0.4242 ± 0.0285	0.3676 ± 0.0360
0.50	0.6158 ± 0.0384	0.5009 ± 0.0590	0.4910 ± 0.0216	0.3676 ± 0.0360

Table 14: Correlation between equivariance defect and orbit OOD relative L^2 across cached N64 Galilean seeds, label fractions, orbit weights, and boost severities. Partial Pearson r residualizes source, method, label fraction, maximum boost, and λ_{orb} .

Scope	Obs.	Pearson r	Spearman ρ	Partial r
All	90	0.851	0.838	0.779
Train/lambda/label	40	0.990	0.990	0.976
OOD severity	50	0.761	0.754	0.516

at low label counts. Burgers and D4 are supporting checks rather than equally strong replications. The five-seed, near-capacity-matched DeepONet reproduction gives the same direction for ID error, orbit OOD error, and defect on the same dataset. Its predictive reductions are 7.2% and 7.0%, compared with 26.6% and 25.1% for FNO, while its 29.6% defect reduction is close to FNO’s 29.8%. This is one fixed-grid, non-FNO branch-trunk operator check, not evidence across operator families and equations. The normalized molecular MLP experiment is a secondary cross-domain mechanism check, but it is neither a neural operator nor a competitive atomistic architecture. Conversely, the D4 G-FNO illustrates backbone dependence: a hard equivariant architecture already saturates the same-group defect and leaves no meaningful consistency signal. Further non-Fourier operator families, resolutions, and datasets remain future work.

The tangent and raw finite-transform controls isolate only the tested coefficients and schedules. They distinguish explicit identity-tangent regularization from finite group evaluations, and test the leading-order role of step normalization, but they do not exhaustively tune every objective. Likewise, matching batch-level forward evaluations does not match backward passes or all transformation overhead. LOCO adds training work; its deployment claim is only that it adds no inference-time module, parameter, or additional operator call.

Finally, the method is only as sound as the action supplied to it. Exact use requires closure of the equation, geometry, coefficients, forcing and boundary data, state representation, and observation map under the transformation. The Dirichlet heat diagnostic shows that a spatial mask can localize an approximate constraint but cannot restore a broken global symmetry. Applying LOCO to a plausible but invalid action may degrade predictive accuracy, so solver-level identity checks and boundary diagnostics should precede long training runs.

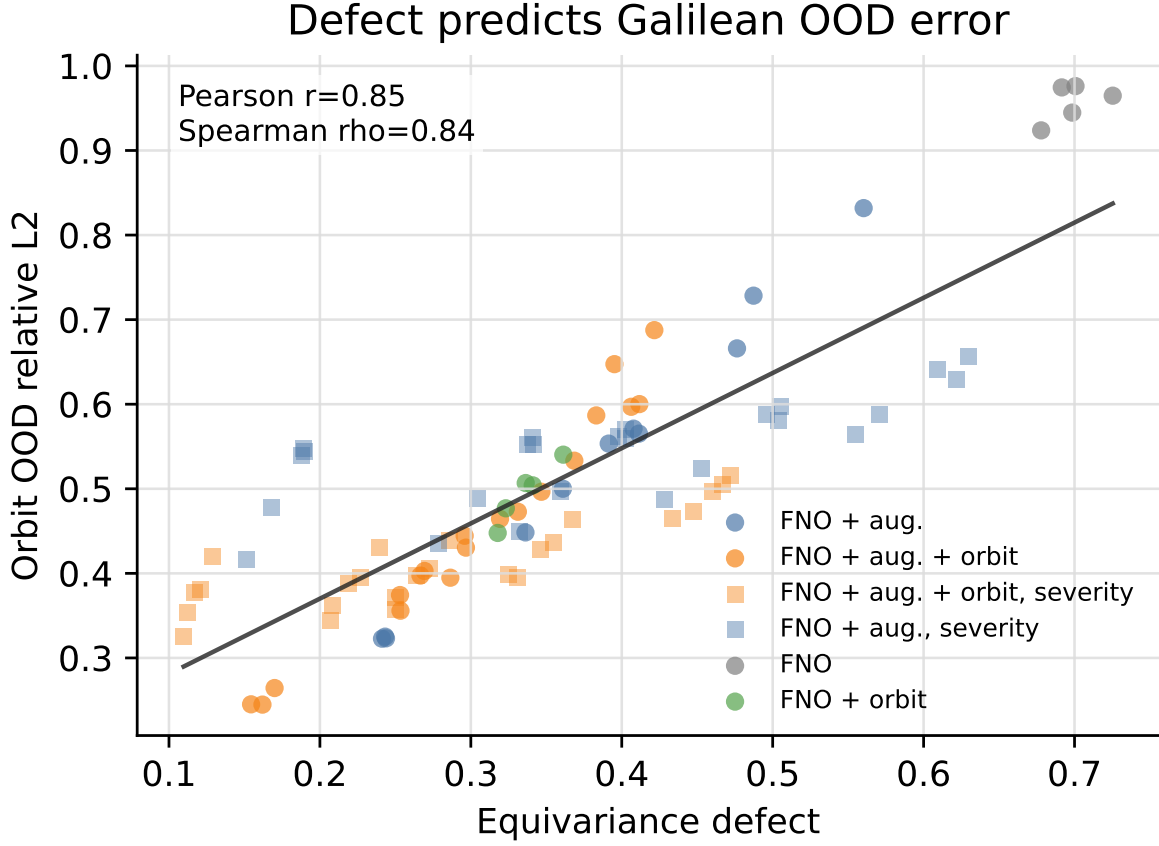


Figure 8: Association between equivariance defect and Galilean orbit OOD error across cached N64 seeds, label fractions, orbit weights, and boost severities. Circles are trained-model evaluations and squares are severity evaluations. Shared datasets and pipelines make this a diagnostic association rather than a causal or out-of-sample prediction result.

Table 15: Orbit-weight ablation at 2% labels. $\lambda_{\text{orb}} = 0.10$ is used for the broader experiments; $\lambda_{\text{orb}} = 0.30$ is the strongest among the tested values.

λ_{orb}	ID L^2	Orbit OOD L^2	Eq. defect	Seeds
0.10	0.3676 ± 0.0360	0.3950 ± 0.0334	0.2677 ± 0.0174	5
0.20	0.3291 ± 0.0318	0.3564 ± 0.0304	0.2268 ± 0.0165	5
0.30	0.3149 ± 0.0298	0.3416 ± 0.0289	0.2044 ± 0.0160	5

Table 16: Relative reductions of augmentation plus orbit consistency over forward-evaluation-matched augmentation across Galilean OOD severities at 2% labels.

Max boost	OOD reduction	Defect reduction
0.10	26.4%	33.6%
0.20	25.8%	31.3%
0.25	25.3%	30.2%
0.35	23.6%	27.8%
0.50	20.3%	23.6%

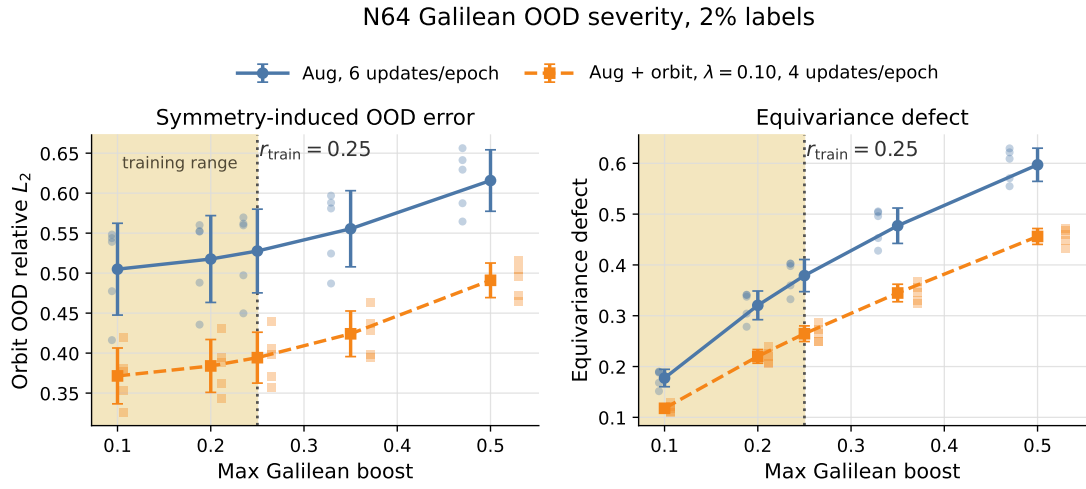


Figure 9: OOD severity at 2% labels. The dashed line marks the componentwise training radius. Augmentation plus orbit consistency remains better than forward-evaluation-matched augmentation from below the training radius through twice the training radius.

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